

# Limits of Binary Representation

## 1. Largest positive number with 8 bits in unsigned binary:

Largest value =  $11111111_2$

Compute it:

$$11111111_2 = 2^7 + 2^6 + \dots + 2^0 = 255$$

**Largest unsigned 8-bit value = 255**

## 2. Largest positive number with 8 bits in signed binary (two's complement):

In two's complement:

Binary:  $01111111_2$

$$01111111_2 = 127$$

**Largest signed 8-bit positive value = 127**

## 3. Smallest negative number with 8 bits in signed binary (two's complement):

In two's complement:

Binary:  $10000000_2$

Value:

$$-2^7 = -128$$

**Smallest signed 8-bit negative value = -128**